



VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JNANA SANGAMA”, BELAGAVI - 590 018

A MINI PROJECT REPORT

on

**“CAMPUS COMPLAINT AND MAINTENANCE
MANAGEMENT SYSTEM”**

Submitted by

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In partial fulfillment of the requirements for the IV semester

DATABASE MANAGEMENT SYSTEMS

of

BACHELOR OF ENGINEERING

in

**COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)**

Under the Guidance of

Mrs. Sangeetha MS

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SAHYADRI

College of Engineering & Management

An Autonomous Institution

MANGALURU

2025–26

SAHYADRI
College of Engineering & Management
An Autonomous Institution
MANGALURU

Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning)



CERTIFICATE

This is to certify that the Mini Project entitled “**CAMPUS COMPLAINT AND MAINTENANCE MANAGEMENT SYSTEM**” has been carried out by **Chaitrika** (4SF24CI042), and **Rihaa Fathima** (4SF24CI133), the bonafide students of **Sahyadri College of Engineering & Management**, in partial fulfillment of the requirements for the IV semester **DATABASE MANAGEMENT SYSTEMS (BAI402G)** of Bachelor of Engineering in Computer Science and Engineering (AI&ML) of Visvesvaraya Technological University, Belagavi during the year 2025–26.

It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of mini project work.

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DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “**CAMPUS COMPLAINT AND MAINTENANCE MANAGEMENT SYSTEM**” has been carried out by us at **Sahyadri College of Engineering & Management, MANGALURU** under the supervision of **Mrs. Sangeetha MS** as the part of the IV semester **DATABASE MANAGEMENT SYSTEMS (BAI402G)** of Bachelor of Engineering in Computer Science and Engineering (AI&ML).

This report has not been submitted to this or any other University.

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Abstract

The Campus Complaint and Maintenance Management System is a database-driven application developed to simplify complaint registration and maintenance handling within a college campus. In many institutions, maintenance complaints are handled manually, making it difficult to track issues, maintain records, and ensure timely resolution. This project provides a centralized and organized platform where students can register complaints related to electrical faults, plumbing problems, classroom maintenance, and network issues.

The system provides separate modules for students, administrators, and technicians. Students can register complaints and track their status, administrators can assign complaints to technicians, and technicians can update complaint progress. The system also maintains complaint history using status logs and allows students to provide feedback after issue resolution.

The project is implemented using Java Swing for the frontend and Oracle Database for backend data management with JDBC connectivity. Database concepts such as ER modeling, relational schema mapping, normalization, triggers, and stored procedures are used in the system implementation. The application improves transparency, reduces paperwork, and enables efficient complaint management within the campus environment.

Acknowledgement

It is with great satisfaction and euphoria that we are submitting the Mini Project Report on “**CAMPUS COMPLAINT AND MAINTENANCE MANAGEMENT SYSTEM**”. We have completed it as a part of the IV semester **DATABASE MANAGEMENT SYSTEMS (BAI402G)** of Bachelor of Engineering in Computer Science and Engineering (AI&ML) of Visvesvaraya Technological University, Belagavi.

We are profoundly indebted to our guide, **Mrs. Sangeetha MS**, Assistant Professor, Department of Computer Science and Engineering (AI&ML) for innumerable acts of timely advice, encouragement and we sincerely express our gratitude.

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Contents

Abstract	i
Acknowledgement	ii
1 Introduction	1
2 Requirements Specification	2
2.1 Hardware Specification	2
2.2 Software Specification	2
3 System Design	3
3.1 ER Diagram	3
3.2 Assumptions	4
3.3 Mapping From ER Diagram to Schema Diagram	4
3.4 Schema Diagram	4
3.5 Normalization	5
4 Implementation	7
4.1 Tables Used	8
5 Results and Discussion	10
6 Conclusion and Future Work	12
6.1 Conclusion	12
6.2 Future Work	12
References	13

List of Figures

3.1	Entity Relationship Diagram for Campus Complaint and Maintenance Management System.	3
3.2	Schema Diagram of Campus Complaint and Maintenance Management System	5
5.1	Student Dashboard	10
5.2	Admin Dashboard	11
5.3	Technician Dashboard	11

List of Tables

4.1	Work Flow	8
4.2	Test case	9

Chapter 1

Introduction

Information systems play an important role in improving the efficiency of institutional management and communication. Educational institutions often face difficulties in handling campus maintenance complaints due to manual processes, delayed responses, and lack of proper tracking systems. Complaints related to electrical faults, plumbing issues, classroom maintenance, and network problems are commonly reported by students and staff. Maintaining records manually makes the process time-consuming and inefficient.

The Campus Complaint and Maintenance Management System is developed to provide an organized and centralized platform for handling maintenance complaints within the campus. The system allows students to register complaints through a user-friendly interface, while administrators can monitor complaints and assign technicians based on specialization and availability. Technicians can update complaint status and maintain progress records until resolution.

The project uses Java Swing for frontend development and Oracle Database for backend data management with JDBC connectivity. The application demonstrates the practical implementation of database concepts such as relational database design, ER modeling, normalization, triggers, stored procedures, and schema mapping. The main objective of the system is to improve transparency, reduce paperwork, maintain complaint history, and ensure efficient maintenance management within the institution.

Chapter 2

Requirements Specification

2.1 Hardware Specification

- Processor : Intel Core i3/i5 or higher
- RAM : 8 GB
- Hard Disk : 256 GB or higher
- Input Devices : Keyboard and Mouse
- Output Device : Monitor

2.2 Software Specification

- Frontend : Java Swing
- Backend : Oracle Database
- Connectivity : JDBC
- Programming Language : Java
- IDE : Visual Studio Code
- Operating System : Windows 10/11

Chapter 3

System Design

3.1 ER Diagram

The ER Diagram of the Campus Complaint and Maintenance Management System represents the relationship between different entities involved in complaint management. The main entities used in the system are USER, CATEGORY, TECHNICIAN, COMPLAINT, ASSIGNMENT, STATUS_LOG, and FEEDBACK.

A user can raise multiple complaints, and each complaint belongs to a specific category. Complaints are assigned to technicians through the ASSIGNMENT entity. The STATUS_LOG entity stores all complaint status updates, while the FEEDBACK entity stores user feedback after complaint resolution. The ER diagram helps in understanding entity relationships and forms the basis for relational

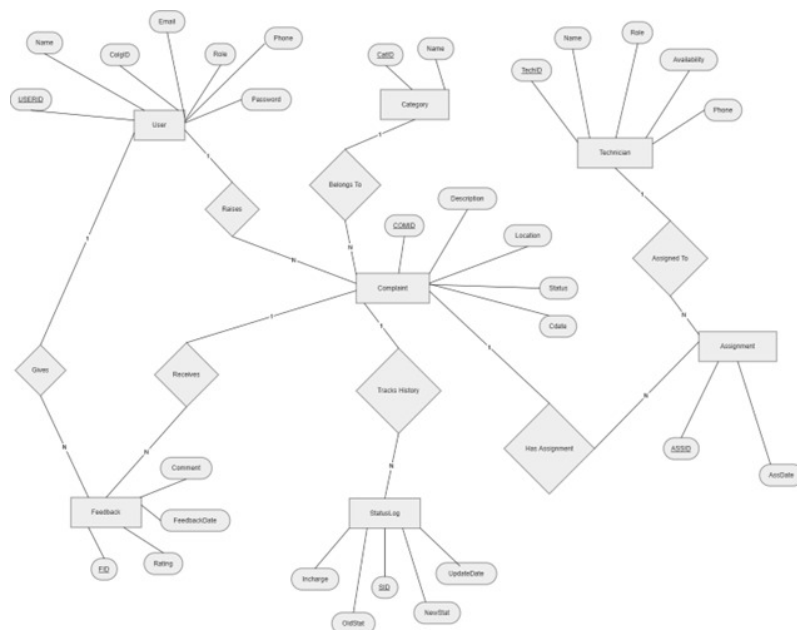


Figure 3.1: Entity Relationship Diagram for Campus Complaint and Maintenance Management System.

3.2 Assumptions

The following assumptions are considered during the development of the system:

- Only registered users can raise complaints.
- One complaint belongs to one category only.
- One technician can handle multiple complaints.
- A complaint can be assigned to only one technician at a time.
- Feedback can be submitted only after complaint resolution.
- Administrators are responsible for assigning technicians.
- Complaint status values are limited to Pending, Assigned, In Progress, and Resolved.

3.3 Mapping From ER Diagram to Schema Diagram

The ER diagram is converted into relational schema using ER-to-relational mapping techniques. Each entity in the ER model is converted into a separate relation with appropriate primary keys and foreign keys.

The USER entity stores student and administrator details. The COMPLAINT table references USER and CATEGORY entities using foreign keys. The ASSIGNMENT table creates the relationship between complaints and technicians. STATUS_LOG maintains complaint history, while FEEDBACK stores user reviews after complaint completion.

The relational schema ensures data consistency, minimizes redundancy, and maintains referential integrity within the database system.

3.4 Schema Diagram

The schema diagram represents the relational structure of the database system and illustrates the relationships between tables using primary and foreign keys.

Relational Schema

- **USER**(user_id PK, name, usn, email, phone, role, password)
- **CATEGORY**(category_id PK, category_name)
- **TECHNICIAN**(technician_id PK, name, phone, specialization, status)

- **COMPLAINT**(complaint_id PK, user_id FK, category_id FK, location, description, complaint_date, status)
- **ASSIGNMENT**(assignment_id PK, complaint_id FK, technician_id FK, assigned_date)
- **STATUS_LOG**(log_id PK, complaint_id FK, old_status, new_status, incharge, changed_date)
- **FEEDBACK**(feedback_id PK, complaint_id FK, user_id FK, rating, comments, feedback_date)

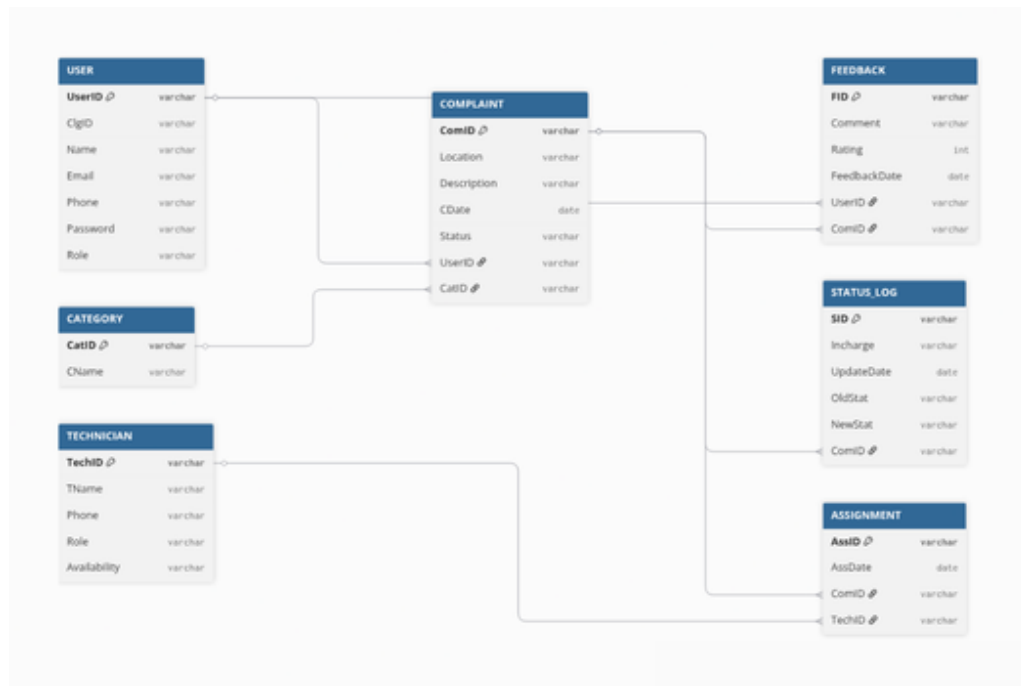


Figure 3.2: Schema Diagram of Campus Complaint and Maintenance Management System

3.5 Normalization

Normalization is performed to reduce data redundancy and improve data consistency within the database.

- **First Normal Form (1NF):**
All attributes in each table contain atomic values and there are no repeating groups.
- **Second Normal Form (2NF):**
All non-key attributes are fully dependent on the primary key. Partial dependency is removed from the relations.
- **Third Normal Form (3NF):**
Transitive dependency is eliminated and all non-key attributes depend only on the primary key. This improves database efficiency and minimizes redundancy.

The database design for the Campus Complaint and Maintenance Management System satisfies normalization requirements up to Third Normal Form (3NF).

Chapter 4

Implementation

The implementation phase involves developing the frontend application using Java Swing and connecting it with Oracle Database using JDBC connectivity. The system is divided into multiple modules for students, administrators, and technicians.

The student module allows users to register complaints, track complaint status, and provide feedback after resolution. The administrator module enables complaint monitoring and technician assignment. The technician module allows technicians to view assigned complaints and update their status.

Triggers and stored procedures are implemented to automate complaint management operations. A trigger automatically inserts records into STATUS_LOG whenever complaint status changes. A stored procedure is used for technician assignment and complaint status management.

Modules Implemented

- Login and Registration Module
- Complaint Registration Module
- Technician Assignment Module
- Complaint Tracking Module
- Status Update Module
- Feedback Management Module

Pseudocode for Complaint Registration

1. User logs into the system

2. User selects complaint category
3. User enters location and description
4. System stores complaint in COMPLAINT table
5. Complaint status is set to Pending
6. Status entry is inserted into STATUS_LOG

Pseudocode for Technician Assignment

1. Administrator views pending complaints
2. Administrator selects technician
3. System updates ASSIGNMENT table
4. Complaint status changes to Assigned
5. Trigger updates STATUS_LOG automatically

4.1 Tables Used

- **USERS:** (UserID, Name, ColgID, Email, Role, Phone, Password)
- **CATEGORY:** (CatID, Name)
- **COMPLAINT:** (COMID, Description, Location, CDate, Status)
- **TECHNICIAN:** (TechID, Name, Role, Availability, Phone)
- **ASSIGNMENT:** (ASSID, AssDate)
- **STATUS_LOG:** (SID, OldStat, NewStat, UpdateDate, Incharge)
- **FEEDBACK:** (FID, Rating, Comment, FeedbackDate)

Sl. No.	Work	Duration(in Weeks)
1	Requirement Analysis & UI Planning	1
2	Database Design & ER Modeling	2
3	Frontend Development (Java Swing)	3
4	Backend Connectivity & Triggers (Oracle)	2
5	System Integration & Testing	1

Table 4.1: Work Flow

TC#	Description	Expected Result	Actual Result	Status
TC-1	Student raises complaint	DB Status 'PENDING'	'PENDING' displayed	Pass
TC-2	Admin assigns technician	Status updates to 'ASSIGNED'	Assignment created	Pass
TC-3	Trigger fires on update	Row added to STATUS_LOG	Audit trail created	Pass

Table 4.2: Test case

Chapter 5

Results and Discussion

The implementation of the Campus Complaint and Maintenance Management System yielded a fully functional, multi-tier desktop application that successfully digitizes the previously manual workflow. Testing across different user roles confirmed that the system correctly enforces data access boundaries and automates routine database tasks.

Key Results Achieved:

- **Seamless Issue Logging:** The Student Dashboard allows for real-time complaint registration, instantly writing the data to the Oracle backend without lag or data loss.
- **Efficient Task Routing:** The Admin Dashboard effectively pulls dynamic lists of pending complaints and available technicians, executing the assignment stored procedure flawlessly.
- **Automated Auditing:** Database verification confirmed that the `trg_status_log` trigger fires precisely upon every status update, successfully creating an immutable audit trail without requiring additional frontend Java code.

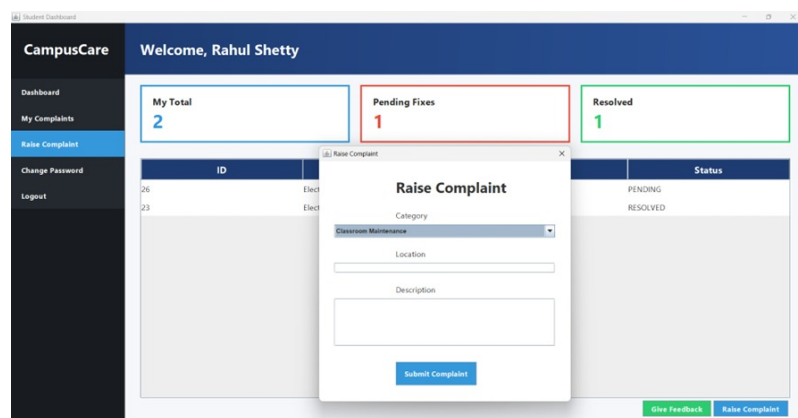


Figure 5.1: Student Dashboard

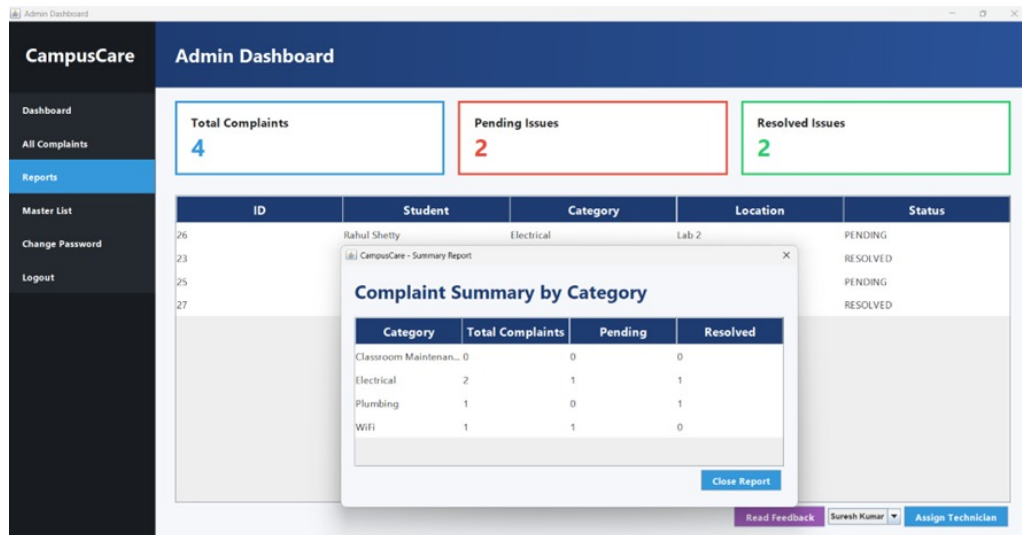


Figure 5.2: Admin Dashboard

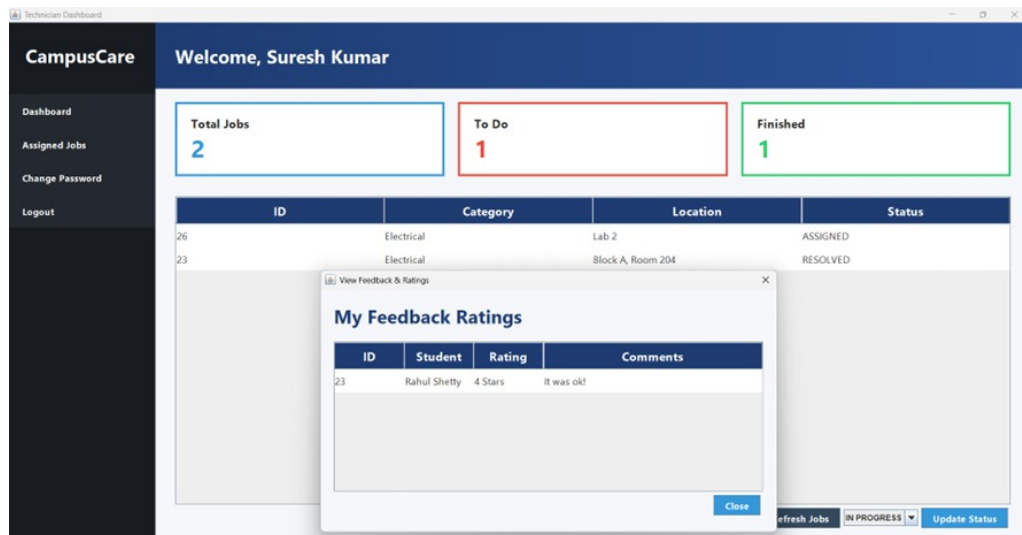


Figure 5.3: Technician Dashboard

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The Campus Complaint and Maintenance Management System successfully bridges the communication gap between students, administrators, and maintenance staff. By transitioning from a physical, paper-based reporting system to a centralized digital platform, the project drastically reduces the response time for resolving campus infrastructure issues. The development of this system demonstrated the successful practical application of core Database Management Systems principles. Utilizing Oracle Database allowed for robust relational schema design and the enforcement of data integrity through normalization. Furthermore, embedding business logic directly into the database layer via automated triggers and stored procedures proved highly effective in reducing frontend complexity and ensuring a foolproof audit trail. Ultimately, the system improves transparency, enhances accountability, and streamlines the overall maintenance lifecycle.

6.2 Future Work

While the current desktop-based system meets the core requirements, a few practical enhancements could be implemented to improve the daily workflow:

- **Automated Notifications:** Extending the backend logic to integrate with a basic email API, sending automatic alerts to technicians the moment a new job is assigned to them.
- **Visual Analytics Dashboard:** Upgrading the Admin module to include graphical charts to help the college track monthly maintenance trends.
- **QR Code Integration:** Generating location-specific QR codes for classrooms and labs. Students could scan these to quickly pull up the complaint registration form with the location data already pre-filled.

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